

UNDERSTANDING PLANETARY ORBITS

THE MOVEMENTS OF THE PLANETS CAN BE DESCRIBED WITH THREE LAWS AND EXPLAINED BY GRAVITY

The eight planets of the Solar System, as well as millions of smaller bodies such as comets and asteroids, travel around the Sun in closed loops called orbits. What keeps these objects on their curved trajectories is the same force that makes things fall to the ground on Earth: gravity.

For centuries it was generally believed that the Earth was the center of the Universe, with the Sun, Moon, planets, and stars rotating around it. However, this geocentric model could not satisfactorily account for planetary orbits, and in 1543 Danish astronomer Nicolaus Copernicus (1473–1543) proposed his heliocentric (Sun-centered) model, with the planets moving around the Sun in circular orbits (see 1543).

KEPLER'S LAWS

In the early 1600s, German astronomer Johannes Kepler (1571–1630) used observations of planetary movements to try to prove Copernicus right. However, he could make the observations fit a heliocentric system only if the orbits were not circles but ellipses, with the Sun at one focus (see below left). This fact became the first of Kepler's three laws of planetary motion. His second law (see below right) relates to the way a planet's speed changes during its orbit, and his third law (see opposite) concerns the relationship between a planet's distance from the Sun and how long it takes to complete each orbit (its orbital period).

GRAVITATIONAL FORCE

Kepler had no idea why orbits should be elliptical. The answer came after his death from English scientist Isaac Newton (1642–1727), who

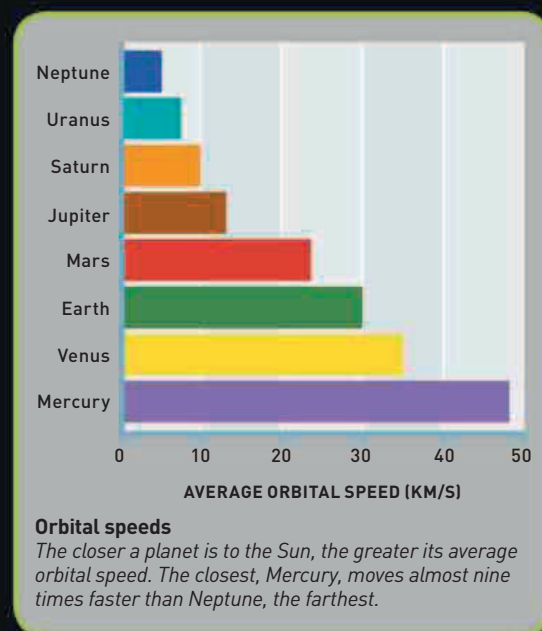


JOHANNES KEPLER

Kepler was an assistant to the great Danish astronomer Tycho Brahe. He used Brahe's observations of the planets when formulating his laws of motion.

suggested that the same force that makes objects fall to the ground on

Earth—gravity—might also be keeping the Moon in orbit around our planet. Newton realized that the force of gravity is weaker the further you are from the center of the Earth, and he proposed that gravity weakened in direct proportion to the square of the distance. When he applied this to



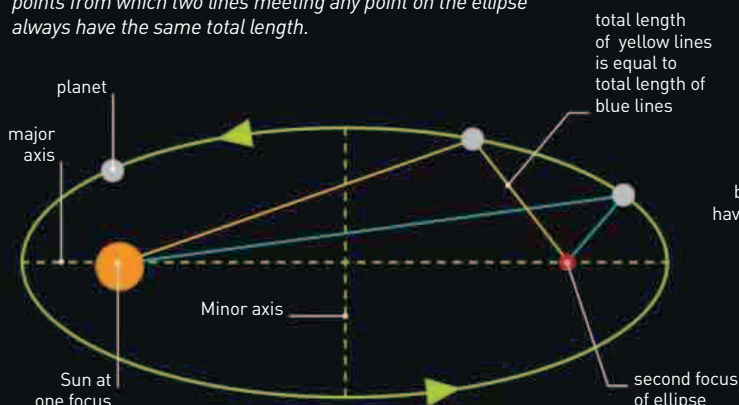
the Moon, he was able to work out the Moon's orbital period. This allowed him to formulate his universal law of gravitation (see 1687) and to realize that gravity must also be responsible for keeping the planets in orbit around the Sun.

92,955,778

THE AVERAGE DISTANCE IN MILES BETWEEN THE SUN AND THE EARTH

ELLIPTICAL ORBITS

Kepler's first law states that every planet's orbit is an ellipse with the Sun at one focus. An ellipse has two foci; they are the points from which two lines meeting any point on the ellipse always have the same total length.



SPEED AND DISTANCE

The second of Kepler's laws states that an imaginary line joining a planet to the Sun sweeps across equal areas in equal times. This takes into account the fact that a planet moves faster when it is closer to the Sun and slower when it is farther away.

